

Consider the following topology on $X = \{a, b, c, d, e\}$.

$$\mathcal{T} = \{\emptyset, X, \{a\}, \{a, b\}, \{a, c, d\}, \{a, b, c, d\}, \{a, b, e\}\}$$

The member of the relative topology on $A = \{a, b\}$ is

Select the correct option

 Reload Math Ea

☒

$\{a\}$

☐

$\{b\}$

☐

$\{c\}$

☐

none

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1



BC200416023: ARSHAD NAWAZ

MTH634 - Topology (Quiz No. 2)

Time Left 86
sec(s)

Quiz Start Time: 09:27 PM, 06 June 2023

Question # 4 of 10 (Start time: 09:37:26 PM, 06 June 2023)

If A is a subset of a topological space X , then

Total Marks: 1

Select the correct option

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☐ $\text{Ext}(A) \subset A^c$ ☐ $\text{Ext}(A) \subset A$

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Question # 4 of 10 (Start time: 09:50:42 PM, 06 June 2023)

If A is a subset of a topological space X , then

Select the correct option

☐	$Ext(A) \cup Int(A) = \phi$
☐	$Ext(A) \cap Int(A) = \phi$

Question # 5 of 10 (Start time: 09:51:17 PM, 06 June 2023)

If A is a subset of a topological space X , then

Select the correct option



$\text{Ext}(A)$ is the smallest open subset of A° .



$\text{Ext}(A)$ is the largest open subset of A° .



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Qaida Nuraniyah....mp4



Roma and Diana i....mp4



Let $X = \{a, b, c, d\}$ with the topology $\tau = \{\emptyset, \{a\}, \{a, b\}, \{a, c, d\}, X\}$, then which of the following represents a subspace topology relative to $A = \{a, b, c\}$?

Select the correct option

☐ $\tau_A = \{\emptyset, \{a\}, \{a, b\}, \{a, c\}, A\}$

☐ $\tau_A = \{\emptyset, \{a\}, \{a, c, d\}, A\}$

☐ $\tau_A = \{\emptyset, \{a\}, \{a, c\}, A\}$

☐ $\tau_A = \{\emptyset, \{a\}, \{a, b\}, A\}$

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Qaida Nuraniyah....mp4

Roma and Diana i....mp4

TuTiTu



The intersection of two neighborhoods of a point is also its neighborhood in a topological space.

Select the correct option

True



False



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Roma and Diana i....mp4



Let X be a set, and let $B = \{\{x\} : x \in X\}$. Which of the following statement is true?

Select the correct option

☐ B is a basis for the topology on X



☐ B is not a basis for the topology on X

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Question # 9 of 10 (Start time: 10:15:27 PM, 06 June 2023)

Total Marks: 1

Consider the following topology on $X = \{a, b, c, d, e\}$:

$$\tau = \{\emptyset, X, \{a\}, \{a, b\}, \{a, c, d\}, \{a, b, c, d\}, \{a, b, e\}\}$$

The neighborhood of the point e is

Select the correct option

[Reload Math Equations](#)☐

none

☒

$\{a, b, c, e\}$

☐

$\{a, b, c\}$

☐

$\{a, b\}$

[Click to Save Answer & Move to Next Question](#)

Question # 5 of 10 (Start time: 03:53:02 PM, 06 June 2023)

The set of all singletons forms a basis for topology.

Select the correct option

☐ lower limit

☐ discrete

☐ co finite

☐ indiscrete

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Click to Save Answer & Move to Next



36°C Sunny



DELL

The boundary of the four intervals $[a, b]$, (a, b) , $(a, b]$, $[a, b)$ is

Select the correct option

(a, b)



$\{a, b\}$



Question # 3 of 10 (Start time: 03:51:21 PM, 06 June 2023)

Interior of a set A (denoted by A°) which is a subset of a topological space X is the of all subsets of X contained in A .

Select the correct option

- | | |
|----------------------------------|----------------------|
| ☒ | union, open |
| ☐ | union, closed |
| ☐ | intersection, closed |
| ☐ | intersection, open |



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Question # 9 of 10 (Start time: 03:50:45 PM, 06 June 2023)

Let $X = \{a, b, c\}$. Then which of the following is not a basis for any topology on X ?

Select the correct option

☐ $B = \{\{a, b\}, \{b, c\}\}$

☐ $B = \{\{a, b\}, \{b\}, \{b, c\}\}$

Question # 9 of 10 (Start time: 03:45:30 PM, 06 June 2023)

Total Marks: 1

Let A be a non empty proper subset of an indiscrete topological space X . The boundary of A is

Select the correct option

☐ $\emptyset$ ☒ X [Click to Save Answer & Move to Next Question](#)

Exterior point of a set is the point of its complement.

Select the correct option



boundary



exterior



limit



interior



Click to Save Answer & Move to

Let $\mathbb{R}$ be real number. The following statement is true.

Select the correct option



The class δ of all infinite open intervals is a subbase for $\mathbb{R}$.

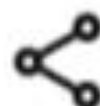


The class δ of all infinite close intervals is a subbase for $\mathbb{R}$.



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The following collection is a basis on $\mathbb{R}^2$.

Select the correct option



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$$B = \{(a, b) * (c, d) \in \mathbb{R}^2 : a < b, c < d\}$$



$$B = \{(a, b) \subseteq \mathbb{R} : a < b\}$$

Click to Save Answer & Move to Next

Question # 8 of 10 (Start time: 03:44:40 PM, 06 June 2023)

Total Marks: 1

Let X be nonempty set, and let $B = \{\{x\}: x \in X\}$. If $\mathcal{T}$ is a topology on X and B is a subset of $\mathcal{T}$, then

Select the correct option



Reload Math Equations

☒ $\mathcal{T}$ is the discrete topology on X .☐ $\mathcal{T}$ is the indiscrete topology on X .

Click to Save Answer & Move to Next Question



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The neighborhood system of a point p in an indiscrete space X is

Select the correct option

Reload



p



X



none



$\emptyset$

Click to Save Answer & Move to



Question # 6 of 10 (Start time: 03:44:07 PM, 06 June 2023)

Total

Basis for indiscrete topology (ϕ, X) is_____

Select the correct option

Reload Math Eq



empty set



$\{0\}$



$\{\phi, X\}$



$\{x\}$

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BC190407239: IMTIAZ AKBAR

Time Left

87

sec(s)

MTH634 - Topology (Quiz No. 2)

Quiz Start Time: 03:38 PM, 06 June 2023

Question # 7 of 10 (Start time: 03:43:55 PM, 06 June 2023)

Total Marks: 1

The set of all singletons forms a basis for topology.

Select the correct option



discrete



lower limit

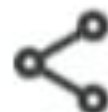


indiscrete



co finite

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Question # 6 of 10 (Start time: 08:43:19 PM, 06 June 2023)

If A is a subset of a topological space X , then

Select the correct option

$\text{Ext}(A)$ is the largest open subset of A^c .



$\text{Ext}(A)$ is the smallest open subset of A^c .



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MTH634 - Topology (Quiz No. 2)

Question # 2 of 10 (Start time: 08:43:13 PM, 06 June 2023)

If A is a subset of a topological space X , then A is open if and only if



Select the correct option



Relo



none



$$A \cap bd(A) \neq \phi$$



$$A \cap bd(A) = A$$



$$A \cap bd(A) = \phi$$

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